Advanced Optimization

Chapter 6 — Quasi-Newton Methods

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Reading: Nocedal & Wright, *Numerical Optimization*, 2nd ed., Chapter 6

Quasi-Newton methods

- Mid 1950s, W. C. Davidon (Argonne National Laboratory): the first quasi-Newton algorithm, developed to accelerate a coordinate descent calculation.
- Fletcher and Powell: much faster and more reliable than the existing methods. Davidon's paper was not accepted for publication; it appeared in 1991.

The idea

Like steepest descent, use only the gradient at each iterate. Measure the changes in gradients and build from them a model of f that is good enough to give superlinear convergence.

No second derivatives are needed. This chapter: small and medium-sized problems.

The quadratic model and the iteration

Model of f at the current iterate x_k:

$$m_k(p) = f_k + \nabla f_k^T p + \tfrac{1}{2} p^T B_k p. \tag{6.1}$$

B_k is an $n \times n$ symmetric positive definite matrix, updated at every iteration. The minimizer of m_k is the search direction:

$$p_k = -B_k^{-1} \nabla f_k \quad (6.2), \qquad\qquad x_{k+1} = x_k + \alpha_k p_k \quad (6.3),$$

where α_k satisfies the Wolfe conditions (3.6).

- This is the line search Newton method with B_k in place of the true Hessian.
- Direction from the model m_k; step length from f itself (Wolfe).
- B_k is updated using the curvature measured during the most recent step.

The secant equation

New model at x_{k+1}: $\quad m_{k+1}(p) = f_{k+1} + \nabla f_{k+1}^T p + \frac{1}{2} p^T B_{k+1} p.$

Requirement: ∇m_{k+1} matches ∇f at the latest two iterates x_k and x_{k+1}.
At x_{k+1}: $\nabla m_{k+1}(0) = \nabla f_{k+1}$, automatically. At x_k:

$$\nabla m_{k+1}(-\alpha_k p_k) = \nabla f_{k+1} - \alpha_k B_{k+1} p_k = \nabla f_k \implies B_{k+1}\alpha_k p_k = \nabla f_{k+1} - \nabla f_k. \quad (6.4)$$

Secant equation

$$s_k = x_{k+1} - x_k = \alpha_k p_k, \qquad y_k = \nabla f_{k+1} - \nabla f_k, \quad (6.5)$$

$$B_{k+1} s_k = y_k. \quad (6.6)$$

The curvature condition

Premultiply (6.6) by s_k^T: a positive definite B_{k+1} can map s_k into y_k only if

$$s_k^T y_k > 0. \tag{6.7}$$

- f strongly convex: (6.7) holds for any two points x_k, x_{k+1} (Exercise 6.1).
- f nonconvex: (6.7) must be enforced by the line search.

The Wolfe conditions (3.6), or strong Wolfe (3.7), guarantee (6.7)

From (6.5) and (3.6b), $\nabla f_{k+1}^T s_k \geq c_2 \nabla f_k^T s_k$, and therefore

$$y_k^T s_k \;\geq\; (c_2 - 1)\, \alpha_k \nabla f_k^T p_k. \tag{6.8}$$

Since $c_2 < 1$ and p_k is a descent direction, the right-hand side is positive.

Determining B_{k+1} uniquely

When (6.7) holds, the secant equation (6.6) has infinitely many solutions: a symmetric matrix has $n(n+1)/2$ degrees of freedom, and the secant equation imposes n conditions.

Additional condition: among all symmetric matrices satisfying the secant equation, B_{k+1} is closest to the current matrix B_k:

$$\min_{B} \quad \|B - B_k\| \tag{6.9a}$$

$$\text{subject to} \quad B = B^T, \qquad Bs_k = y_k. \tag{6.9b}$$

Each matrix norm in (6.9a) gives a different quasi-Newton method.

The weighted Frobenius norm

$$\|A\|_W \equiv \left\| W^{1/2} A W^{1/2} \right\|_F, \qquad \|C\|_F^2 = \sum_{i=1}^n \sum_{j=1}^n c_{ij}^2. \tag{6.10}$$

W can be *any* matrix satisfying $Wy_k = s_k$. For concreteness, $W = \bar{G}_k^{-1}$, where $\bar{G}_k$ is the **average Hessian**

$$\bar{G}_k = \Big[\int_0^1 \nabla^2 f(x_k + \tau \alpha_k p_k)\, d\tau \Big]. \tag{6.11}$$

By Taylor's theorem (Theorem 2.1), $\quad y_k = \bar{G}_k \alpha_k p_k = \bar{G}_k s_k$ (6.12).

With this W the norm is non-dimensional: the solution of (6.9) does not depend on the units of the problem. (B has units $[f]/[x]^2$ and W has units $[x]^2/[f]$, so $W^{1/2}(B - B_k)W^{1/2}$ has no units.)

Why a weighted norm? (Not in the book.)

$\|\cdot\|_W$ is an error metric that takes the curvature into account.

Example: $\nabla^2 f = \mathrm{diag}(1000, 0.001)$ and $B = \nabla^2 f + I$.

- Direction e_1: curvature 1001 instead of 1000. The step is almost exact.
- Direction e_2: curvature 1.001 instead of 0.001. The step is 1000 times too short.

The Frobenius norm gives both errors the size 1. With $W = \nabla^2 f^{-1}$ they have sizes 10^{-3} and 10^3, in proportion to their effect on the step.

Equivalently: in the variables $z = \bar{G}_k^{1/2} x$ the curvature is I in every direction, and $\|\cdot\|_W$ is the ordinary Frobenius norm there. The solution of (6.9) depends on W only through $Wy_k = s_k$, so $\bar{G}_k$ is never computed.

The DFP update

The unique solution of (6.9) in the norm (6.10), with $\rho_k = 1/(y_k^T s_k)$ (6.14):

$$\text{(DFP)} \qquad B_{k+1} = \big(I - \rho_k y_k s_k^T\big) B_k \big(I - \rho_k s_k y_k^T\big) + \rho_k y_k y_k^T. \tag{6.13}$$

With $H_k = B_k^{-1}$ the direction (6.2) is a matrix–vector product. By the Sherman–Morrison–Woodbury formula (A.28):

$$\text{(DFP)} \qquad H_{k+1} = H_k - \frac{H_k y_k y_k^T H_k}{y_k^T H_k y_k} + \frac{s_k s_k^T}{y_k^T s_k}. \tag{6.15}$$

- Davidon (1959); Fletcher and Powell. Both formulas are rank-two modifications: the matrix is not recomputed from scratch.
- The book states (6.13) without derivation. **Read:** pp. 138–139.

The BFGS update

Apply the same argument to H_k instead of B_k. The secant equation is now $H_{k+1}y_k = s_k$:

$$\min_{H} \quad \|H - H_k\| \tag{6.16a}$$

$$\text{subject to} \quad H = H^T, \qquad Hy_k = s_k. \tag{6.16b}$$

Norm (6.10), now with $Ws_k = y_k$. The unique solution, with ρ_k from (6.14):

$$\text{(BFGS)} \qquad H_{k+1} = \big(I - \rho_k s_k y_k^T\big) H_k \big(I - \rho_k y_k s_k^T\big) + \rho_k s_k s_k^T. \tag{6.17}$$

Broyden, Fletcher, Goldfarb, Shanno. Considered the most effective update.

BFGS in terms of B_k; duality

Applying the Sherman–Morrison–Woodbury formula (A.28) to (6.17):

$$\text{(BFGS)} \qquad B_{k+1} = B_k - \frac{B_k s_k s_k^T B_k}{s_k^T B_k s_k} + \frac{y_k y_k^T}{y_k^T s_k}. \tag{6.19}$$

Duality. The DFP and BFGS formulae are obtained from each other by the interchanges

$$s \leftrightarrow y, \qquad B \leftrightarrow H.$$

	B_{k+1}	H_{k+1}
DFP	(6.13)	(6.15)
BFGS	(6.19)	(6.17)

B_k itself is needed when it enters a subproblem: a trust-region step (Chapter 4) or the KKT system of SQP (Chapter 18). For a line search, use H_k (Algorithm 6.1).

Algorithm 6.1 (BFGS method)

Given starting point x_0, convergence tolerance $\epsilon > 0$,
inverse Hessian approximation H_0; $k \leftarrow 0$

while $\|\nabla f_k\| > \epsilon$

compute the search direction $p_k = -H_k \nabla f_k$ (6.18)

set $x_{k+1} = x_k + \alpha_k p_k$, where α_k satisfies the Wolfe conditions (3.6)

define $s_k = x_{k+1} - x_k$ and $y_k = \nabla f_{k+1} - \nabla f_k$

compute H_{k+1} by means of (6.17); $k \leftarrow k + 1$

- Cost per iteration: $O(n^2)$ arithmetic operations. No linear system is solved.
- Rate: superlinear. Newton's method is quadratic, but each iteration needs second derivatives and the solution of a linear system.

Numerical example: Rosenbrock's function (2.22)

Starting point $(-1.2, 1)$. Wolfe conditions on the step length in all three methods.

Iterations until $\|\nabla f_k\| \leq 10^{-5}$

steepest descent	5264
BFGS	34
Newton (inexact)	21

Last four values of $\|x_k - x^*\|$

steepest descent	BFGS	Newton
1.827e-04	1.70e-03	3.48e-02
1.826e-04	1.17e-03	1.44e-02
1.824e-04	1.34e-04	1.82e-04
1.823e-04	1.01e-06	1.17e-08

The superlinear rate of BFGS is visible in the last iterations. **Read:** p. 141.

Properties of the BFGS method

Positive definiteness is preserved

Let H_k be positive definite and $y_k^T s_k > 0$. For any nonzero z,

$$z^T H_{k+1} z = w^T H_k w + \rho_k (z^T s_k)^2 \geq 0, \qquad w = z - \rho_k y_k (s_k^T z).$$

Zero only if $s_k^T z = 0$; then $w = z \neq 0$ and the first term is positive. Hence H_{k+1} is positive definite.

Self-correction. **Read:** pp. 141–142.

- If H_k estimates the curvature incorrectly and this slows down the iteration, H_k tends to correct itself within a few steps. DFP is less effective at this.
- The property holds only with an adequate line search (Wolfe).

BFGS versus DFP (Not in the book.)

Iterations; strong Wolfe line search ($c_1 = 10^{-4}$, $c_2 = 0.9$), $\alpha = 1$ tried first.

f	H_0	BFGS	DFP
quadratic, $n = 10$,	$10^{-4}I$	94	> 5000
eigenvalues 1 to 100,	$10^{-2}I$	45	249
$\|\nabla f_k\| < 10^{-8}$	I	11	15
Rosenbrock (2.22), $\|\nabla f_k\| < 10^{-5}$	I	34	84

- DFP changes B minimally, so B_k too large (steps too short) persists; a line search starting at $\alpha = 1$ does not repair this.
- BFGS changes H minimally: H_k too large (steps too long) is fixed by the line search.
- Theorem 6.5 extends to $\phi_k \in [0, 1)$ in (6.32), but not to DFP ($\phi_k = 1$).

Implementation: the line search

- Use the Wolfe conditions (3.6) or the strong Wolfe conditions (3.7), and always try $\alpha_k = 1$ first. This step length will eventually always be accepted (under certain conditions), which gives superlinear convergence.
- A fairly inaccurate line search is more economical in function evaluations. Commonly used: $c_1 = 10^{-4}$, $c_2 = 0.9$.
- Armijo backtracking from $\alpha = 1$ does not guarantee $y_k^T s_k > 0$: a step length greater than 1 may be required.
- Skipping the update ($H_{k+1} = H_k$) when $y_k^T s_k$ is negative or too close to zero is not recommended: the updates may be skipped much too often.

Read: pp. 142–143.

Implementation: the initial matrix H_0

$H_0 = \beta I$: there is no good general strategy for choosing β.

Heuristic: scale H_0 *after* the first step has been computed and *before* the first update:

$$H_0 \leftarrow \frac{y_k^T s_k}{y_k^T y_k} I. \tag{6.20}$$

The factor approximates an eigenvalue of $\nabla^2 f(x_k)^{-1}$, so (6.20) makes the size of H_0 similar to that of $\nabla^2 f(x_0)^{-1}$. **Read:** (6.21), p. 143.

(Not in the book.) With (6.20), the number of BFGS iterations hardly depends on the initial β: on the quadratic in the BFGS-versus-DFP table, 43–44 iterations for $\beta = 10^{-4}, 10^{-2}, 1$ (without (6.20): 94, 45, 11). (6.20) removes the choice of β; it does not always reduce the iteration count.

The symmetric rank-one update: derivation

BFGS and DFP are rank-two updates. A rank-one update that keeps symmetry: $B_{k+1} = B_k + \sigma v v^T$, with $\sigma = +1$ or -1. Impose the secant equation $y_k = B_{k+1} s_k$:

$$y_k = B_k s_k + \left[\sigma v^T s_k\right] v. \tag{6.22}$$

The bracket is a scalar, so $v = \delta(y_k - B_k s_k)$ for some scalar δ. Substituting,

$$(y_k - B_k s_k) = \sigma \delta^2 \left[s_k^T (y_k - B_k s_k)\right](y_k - B_k s_k). \tag{6.23}$$

This holds if and only if $\sigma = \operatorname{sign}\left[s_k^T (y_k - B_k s_k)\right]$, $\delta = \pm \left|s_k^T (y_k - B_k s_k)\right|^{-1/2}$. Then $\sigma v v^T = \sigma \delta^2 (y_k - B_k s_k)(y_k - B_k s_k)^T$ with $\sigma \delta^2 = 1/\left[(y_k - B_k s_k)^T s_k\right]$: this is (6.24).

The SR1 formulas

The only symmetric rank-one update that satisfies the secant equation:

$$\text{(SR1)} \qquad B_{k+1} = B_k + \frac{(y_k - B_k s_k)(y_k - B_k s_k)^T}{(y_k - B_k s_k)^T s_k}. \tag{6.24}$$

By the Sherman–Morrison formula (A.27):

$$\text{(SR1)} \qquad H_{k+1} = H_k + \frac{(s_k - H_k y_k)(s_k - H_k y_k)^T}{(s_k - H_k y_k)^T y_k}. \tag{6.25}$$

- Even if B_k is positive definite, B_{k+1} may not be. With trust-region methods, the ability to generate indefinite approximations is a chief advantage.
- Main drawback: the denominator can vanish, even for a convex quadratic.

Why SR1 is still of interest

(i) A simple safeguard prevents breakdown and numerical instabilities.

(ii) The matrices generated by the SR1 formula tend to be good approximations to the true Hessian matrix, often better than the BFGS approximations.

(iii) In quasi-Newton methods for constrained problems, and for partially separable functions, it may not be possible to impose the curvature condition $y_k^T s_k > 0$, so BFGS updating is not recommended. In these settings indefinite Hessian approximations are desirable insofar as they reflect indefiniteness in the true Hessian.

B_k may be indefinite, so SR1 is used with a trust region (Algorithm 6.2), not a line search. **Read:** pp. 145–149 (Algorithm 6.2, Theorems 6.1, 6.2).

The skipping rule

Safeguard

Apply the update (6.24) only if

$$\left|s_k^T(y_k - B_k s_k)\right| \;\geq\; r\,\|s_k\|\,\|y_k - B_k s_k\|, \tag{6.26}$$

where $r \in (0,1)$ is a small number, say $r = 10^{-8}$. Otherwise set $B_{k+1} = B_k$.

If $y_k \neq B_k s_k$ but $(y_k - B_k s_k)^T s_k = 0$, no SR1 update exists. Why is skipping acceptable for SR1, but not for BFGS?

- **SR1.** $s_k^T(y_k - B_k s_k) \approx 0$ occurs infrequently. It implies $s_k^T \bar{G} s_k \approx s_k^T B_k s_k$: the curvature of B_k along s_k is already correct.
- **BFGS.** Without Wolfe, $s_k^T y_k > 0$ fails easily; skipping can occur often.

The Broyden class

A family of updates with a scalar parameter ϕ_k:

$$B_{k+1} = B_k - \frac{B_k s_k s_k^T B_k}{s_k^T B_k s_k} + \frac{y_k y_k^T}{y_k^T s_k} + \phi_k \left(s_k^T B_k s_k\right) v_k v_k^T, \tag{6.32}$$

$$v_k = \left[\frac{y_k}{y_k^T s_k} - \frac{B_k s_k}{s_k^T B_k s_k}\right]. \tag{6.33}$$

$\phi_k = 0$: BFGS. $\phi_k = 1$: DFP. Equivalently, $B_{k+1} = (1-\phi_k) B_{k+1}^{\mathrm{BFGS}} + \phi_k B_{k+1}^{\mathrm{DFP}}$. All members satisfy the secant equation (6.6). **Restricted Broyden class:** $\phi_k \in [0, 1]$; positive definiteness is preserved when $s_k^T y_k > 0$. **Read:** Theorems 6.3, 6.4 (quadratic f), pp. 150–152.

What is proved, and what is not

- BFGS and SR1 are remarkably robust in practice. For general nonlinear f, however, global convergence is not established: "it is not yet known if the algorithms enjoy such properties" (p. 153).
- (Not in the book.) For nonconvex f the BFGS iterates can fail to converge: examples of Dai (2002, Wolfe line search) and Mascarenhas (2004, exact line search).
- The analysis assumes that f is convex, or that the iterates satisfy certain properties. Tools: the trace and the determinant of B_k, combined in

$$\psi(B) = \text{trace}(B) - \ln(\det(B)). \tag{6.49}$$

Notation of §6.4: $\|\cdot\|$ is the Euclidean vector or matrix norm; $G(x) = \nabla^2 f(x)$.

Global convergence of BFGS: Theorem 6.5

Assumption 6.1

(i) The objective function f is twice continuously differentiable.

(ii) The level set $\mathcal{L} = \{x \in \mathbb{R}^n \mid f(x) \leq f(x_0)\}$ is convex, and there exist positive constants m and M such that $m\|z\|^2 \leq z^T G(x) z \leq M\|z\|^2$ (6.39) for all $z \in \mathbb{R}^n$ and $x \in \mathcal{L}$.

Theorem 6.5

Let B_0 be any symmetric positive definite initial matrix, and let x_0 be a starting point for which Assumption 6.1 is satisfied. Then the sequence $\{x_k\}$ generated by Algorithm 6.1 (with $\epsilon = 0$) converges to the minimizer x^* of f.

Read: proof of Theorem 6.5, pp. 154–156.

Superlinear convergence of BFGS: Theorem 6.6

Assumption 6.2

The Hessian matrix G is Lipschitz continuous at x^*, that is,

$$\|G(x) - G(x^*)\| \leq L\|x - x^*\|,$$

for all x near x^*, where L is a positive constant.

Theorem 6.6

Suppose that f is twice continuously differentiable and that the iterates generated by the BFGS algorithm converge to a minimizer x^* at which Assumption 6.2 holds. Suppose also that (6.52) holds. Then x_k converges to x^* at a superlinear rate.

Read: proof of Theorem 6.6, pp. 157–160.

Reading Theorem 6.6

- Applies to general nonlinear f, not just convex f; $x_k \to x^*$ is an assumption.
- With $G_* = G(x^*)$, the proof establishes

$$\lim_{k\to\infty} \frac{\|(B_k - G_*)s_k\|}{\|s_k\|} = 0.$$

- This is the Dennis–Moré characterization (3.36). By Theorem 3.6, the unit step length $\alpha_k = 1$ satisfies the Wolfe conditions near the solution, and the rate of convergence is superlinear.
- $B_k \to \nabla^2 f(x^*)$ is not required. It suffices that B_k becomes increasingly accurate along the search directions (p. 47).

Read: SR1: Theorem 6.7, pp. 160–161.

Chapter 6 in one table

	BFGS	DFP	SR1
Update	rank two	rank two	rank one
Obtained from	(6.16), closest H	(6.9), closest B	(6.24)
Positive definite	if $s_k^T y_k > 0$	if $s_k^T y_k > 0$	not guaranteed
Breakdown	no	no	possible; skip by (6.26)
Self-correction	very effective	less effective	—
Framework	line search	line search	trust region
ϕ_k in (6.32)	0	1	may be outside $[0, 1]$
Global convergence	Thm. 6.5 (convex f)	not in Thm. 6.5	Thm. 6.7
Rate	superlinear (Thm. 6.6)	—	$(n+1)$-step superlinear
$B_k \to \nabla^2 f(x^*)$	not required	—	Thm. 6.2

Three results to carry forward

1. **The secant equation $B_{k+1}s_k = y_k$.** Curvature information is obtained from differences of gradients. The update costs $O(n^2)$; no second derivatives and no linear system.

2. **The Wolfe conditions are part of the BFGS method.** They guarantee $s_k^T y_k > 0$, which keeps H_k positive definite and p_k a descent direction. Self-correction also depends on them.

3. **Superlinear convergence does not require $B_k \to \nabla^2 f(x^*)$.** Accuracy along the search directions is enough (BFGS, Theorem 6.6). SR1 does approximate the Hessian (Theorems 6.1, 6.2) and allows indefinite B_k within a trust region.

Proved in the book: Theorems 6.1, 6.5, 6.6.
Stated without proof: Theorems 6.2, 6.3, 6.4, 6.7.

Homework (problems 3–5 are not in the book)

1. Exercise 6.1(a): if f is strongly convex, then (6.7) holds for any x_k and x_{k+1}.
2. Exercise 6.2: the second strong Wolfe condition (3.7b) implies (6.7).
3. By hand. $f(x) = \frac{1}{2}x^T Ax - b^T x$, $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $b = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $x_0 = 0$, $H_0 = I$. Run two iterations of Algorithm 6.1 with exact line searches. Check that $H_1 y_0 = s_0$, $x_2 = x^*$, $H_2 = A^{-1}$.
4. By hand. $B_k = I$, $s_k = (1, 0)^T$, $y_k = (-1, 0)^T$. Compute B_{k+1} from (6.24). Is it positive definite? Why can the BFGS update not be used here?
5. Implement Algorithm 6.1 with the scaling (6.20) and a library Wolfe line search. Minimize Rosenbrock's function (2.22) from $(-1.2, 1)$. Compare the last four values of $\|x_k - x^*\|$ with the table on p. 141.